

Round Supervision for Interactive Whole-Body PET/CT Lesion Segmentation^{*}

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Abstract. Interactive lesion segmentation in whole-body PET/CT requires a model that not only produces an accurate initial delineation but also responds correctly to user guidance. We present our contribution to the autoPET V challenge, which targets FDG- and PSMA-tracer whole-body PET/CT with click-based interaction. Our final submission is a 3D residual-encoder U-Net trained under a *round-supervision* regime: rather than training on guidance derived from the ground truth, each training case is presented over multiple simulated interaction rounds in which corrective scribbles are generated from the model’s own false positives and false negatives. Guidance is encoded as two additional input channels holding a spacing-aware local Euclidean distance transform, giving a four-channel input of CT, PET, foreground guidance and background guidance. On the challenge preliminary test set this single model, without ensembling or test-time augmentation, achieves a Dice score of 0.7702 and a lesion-level F1 of 0.7558. Separately, on a ground-truth-guided baseline, we report controlled ablations over guidance encodings (truncated distance, geodesic, disk, Gaussian) and network capacity under identical training budgets: encoding choice among smoothly decaying variants proved far less important than network capacity, and both effects are reported alongside the split-dependent variation that bounds their interpretation. We additionally report two negative results that may be useful to the community: organ-aware false-positive suppression using anatomical priors, and PET-specific intensity normalisation, both of which degraded performance.

Keywords: autoPET challenge · Interactive segmentation · PET/CT · Lesion segmentation · nnU-Net

1 Introduction

Automated lesion segmentation in whole-body PET/CT supports tumour burden quantification, staging and treatment response assessment, but fully automatic methods remain imperfect in clinical practice. The autoPET V challenge addresses this by framing the task as *interactive* segmentation: an algorithm receives a CT volume, a PET volume and a set of user clicks marking lesion

^{*} Challenge team name: **Volyagers**. Algorithm submitted on grand-challenge as autoPETV EDT+.

and background locations, and must produce a lesion segmentation that can be progressively corrected over multiple rounds of interaction.

The challenge data spans two tracers with markedly different physiological uptake patterns. FDG accumulates in brain, myocardium, bowel and bladder; PSMA accumulates in kidneys, bladder and salivary glands. A model that responds only to PET intensity will therefore produce false positives in healthy tissue, and the distribution of those false positives differs by tracer. A further difficulty is that a substantial fraction of the provided studies contain no lesions at all, so any spurious prediction on those cases is penalised without any opportunity for compensating overlap.

Our contribution addresses the interactive aspect directly. Rather than treating user guidance as an auxiliary input added at inference time, we train the network under a regime in which guidance is generated from the model’s own errors across successive rounds, so that the mapping from correction to improvement is learned rather than assumed. We describe the method and its results, and separately report a set of controlled ablations carried out on a ground-truth-guided baseline, which characterise which design factors matter for this task and which do not.

2 Methods

2.1 Data

Training data. We used only the data provided by the autoPET V challenge: whole-body FDG [4,5] and PSMA [6] PET/CT studies with manually annotated tumour lesions. Our final model was trained on 1611 studies. No external or private data were used, and no pre-trained weights were employed.

Validation data. Ablation comparisons were performed on held-out validation folds of the challenge training data, using the split definitions generated by nnU-Net for each dataset (`splits_final.json`), on a fixed 250-case subset (see Section 3). Compute constraints limited the submitted round-supervised configuration to a *single fixed training/validation split*; it was not cross-validated. No challenge test data were used for training or for tuning any model parameter. Preliminary test set submissions were used only to choose between completed candidate models, as permitted by the challenge protocol.

2.2 Data pre-processing

Preprocessing followed the nnU-Net [1] pipeline. CT intensities were clipped to the 0.5–99.5 percentile range of foreground voxels and z-score normalised using dataset-level statistics (`CTNormalization`); PET intensities were z-score normalised per-image. Volumes were resampled to a common spacing of $3.0 \times 2.04 \times 2.04$ mm.

User clicks were rendered into two additional input channels, one for foreground (lesion) clicks and one for background clicks. Each channel holds a

spacing-aware local Euclidean distance transform: voxels within an anchor radius of a click take value 1, the value decays linearly to 0 at a support radius of 40 mm, and voxels beyond that radius are 0. Because the encoding is computed in physical millimetres rather than voxels, it is invariant to the anisotropic and study-dependent voxel spacing of the input. These two channels are passed through the network unnormalised (`NoNormalization`), preserving the encoding’s absolute scale.

2.3 Algorithm/model

The network is a 3D residual-encoder U-Net (`ResidualEncoderUNet`) [2] with six resolution stages, feature widths [32, 64, 128, 256, 320, 320] and [1, 3, 4, 6, 6, 6] residual blocks per stage. The input is four channels (CT, PET, foreground guidance, background guidance) and the output is a two-class segmentation (background, lesion). The patch size is $112 \times 160 \times 128$ voxels.

The distinguishing element of our method is the training regime, which we refer to as *round supervision*. In a conventional interactive setup, guidance channels are synthesised from the ground truth, so at training time the guidance is always correct and always informative. At inference time, however, guidance arrives as corrections to whatever the model has already predicted, which is a different distribution entirely.

Under round supervision, simulated interaction is applied with probability 0.8 using three interaction rounds. The first round supplies no guidance, so the network must produce a purely automatic segmentation. In each subsequent round, guidance is generated from the model’s current false positive and false negative regions: over-segmented regions ($\hat{y} = 1, y = 0$) yield background scribbles and under-segmented regions ($\hat{y} = 0, y = 1$) yield foreground scribbles. Guidance therefore accumulates across rounds and is drawn from the model’s own error distribution, matching the conditions the model encounters at inference.

2.4 Data post-processing

Predicted lesion components were filtered using a heuristic conservative two-criterion rule. Connected components were computed using 18-connectivity, and a component was removed only when its volume was below 0.35 mL *and* its $\text{SUV}_{\max}$ was below 6.0. Requiring both conditions avoids discarding small but metabolically active lesions, which a volume threshold alone would remove. These thresholds were selected during preliminary development and were not established through a comprehensive held-out threshold sweep; consequently, the effect of this postprocessing should be interpreted cautiously.

2.5 Training and test parameters

Training used the nnU-Net defaults: SGD with Nesterov momentum ($\mu = 0.99$), initial learning rate 10^{-2} with polynomial decay $((1 - t/T)^{0.9})$, batch size 2,

and a compound Dice + cross-entropy loss with deep supervision. The model was trained for 1000 epochs of 250 iterations each. The deployed weights are the epoch-950 snapshot, which was used for the reported preliminary challenge evaluation; the packaged checkpoint is pinned by SHA-256 in the submission repository to prevent inadvertent substitution.

Training-time augmentation followed the nnU-Net default pipeline: random rotation, scaling, Gaussian noise, Gaussian blur, brightness and contrast perturbation, low-resolution simulation, gamma correction and mirroring. Test-time augmentation was disabled at inference.

Training was performed on a single NVIDIA H200 GPU. Reaching the epoch-950 snapshot required approximately 97 hours of wall-clock training, with a median of 364 s per epoch of 250 iterations.

2.6 Github repository

<https://github.com/adkSpence/autoPETV>

The results in this paper correspond to branch `master` at commit `2fc2670`. Model weights are tracked with Git LFS; the deployment checkpoint is verified by SHA-256 in `check_weights.sh`.

3 Results

Our development proceeded along two tracks. The first was a ground-truth-guided interactive baseline, on which we ran controlled ablations to characterise which factors matter for this task. The second was the round-supervised model described in Section 2.3, which was trained separately and became our submission. We report both, and are explicit about which conclusions transfer between them.

3.1 Ablation study on the guided baseline

The ablations were performed on a separate four-channel baseline in which guidance channels were synthesised directly from the ground truth, using a truncated distance-to-click encoding bounded at 10 voxels. This is *not* the encoding used by the submitted model (Section 2.3), and the ablations should therefore be read as characterising the task rather than as a selection procedure for the final configuration.

All runs used a fixed 250-case subset, an identical 90-epoch schedule and identical splits, varying one factor at a time. All numbers in Table 2 are Mean Validation Dice obtained by full sliding-window inference on the held-out fold, not training-time pseudo-Dice.

Two observations follow. First, the two distance-based encodings differ by 0.0007 Dice, indistinguishable at this sample size, while both clearly outperform the compact disk and Gaussian encodings. What appears to matter is the presence of a smooth spatial falloff around each click, not the particular distance

metric used to produce it. Second, network capacity produced the single largest effect of any factor we varied (+0.040 Dice), exceeding the spread across every guidance encoding tested.

Both findings are consistent with the design of the submitted model, which uses a residual-encoder architecture and a smoothly decaying guidance encoding. We stress, however, that the submitted model was trained separately, on a different dataset preparation, with a different encoding and a smaller residual-encoder variant (M rather than L, chosen for memory headroom under the multi-round training regime). These ablations therefore corroborate its design rather than having selected it.

3.2 Validation and challenge results

Compute constraints limited the final round-supervised configuration to a single fixed training/validation split. It was trained on one fold only; no ensembling across folds, seeds or architectures was used in the final submission, and test-time augmentation was disabled. We therefore do not report cross-validation results for the submitted model.

In a separate ground-truth-guided baseline experiment, two independently trained splits differed by 0.058 Dice (0.6714 and 0.7290, 323 validation cases per split). Although this is not cross-validation evidence for the submitted round-supervised model, it illustrates the potential magnitude of split-dependent variation on this dataset: it exceeds the gap between every guidance encoding in Table 2. On that basis we consider the architecture result (+0.040) and the PET normalisation result (−0.065) large enough to be credible, and the 0.0007 gap between the two distance encodings not to be.

On the challenge preliminary test set, our submitted algorithm achieved the results in Table 3.

3.3 Negative results

We report two approaches that did not work, since both are intuitively appealing and may otherwise be re-attempted. Both were evaluated on the guided baseline described above, not on the submitted model.

Organ-aware false-positive suppression. Physiological tracer uptake in bladder, bowel, kidneys, myocardium and liver is a known source of false positives. We used TotalSegmentator [3] to segment these structures on CT and suppressed predicted lesion components exceeding an overlap threshold with them. A pilot on five cases was encouraging: false positive components were approximately 4.5 times more likely to overlap a flagged organ than true positives (15.6% versus 3.5%). Evaluated across all 323 validation cases, however, the method was negative on both Dice and lesion-level F1 at every overlap threshold tested. Precision improved but recall fell faster: the same anatomy that generates false positives also harbours genuine disease, and no purely spatial rule separated the two.

PET-specific intensity normalisation. Replacing per-image z-score normalisation of the PET channel with an SUV-aware scheme reduced Dice from 0.7102 to 0.6457 under otherwise identical conditions.

4 Discussion

Taken together, our results separate two questions that are easily conflated. *Which* smooth guidance encoding is chosen appears to matter little: the two distance-based variants in Table 2 differ by 0.0007 Dice, well inside the fold-to-fold noise we measured. Whether the encoding used at inference *matches* the one used in training, by contrast, matters enormously, as the following deployment failure demonstrates.

This was reinforced by a deployment failure that is worth reporting explicitly. An earlier submission of the same trained weights scored 0.40 Dice on the preliminary test set. The cause was not the model but the inference-time encoding: our click renderer produced a truncated distance transform (value 0 at the click, rising with distance, measured in voxels), whereas the network had been trained on a local support (value 1 at the click, decaying to 0 at 40 mm, measured in millimetres). The two encodings are approximately inverses of one another, so the network received guidance whose polarity, scale and units were all wrong. The corrected deployment uses a spacing-aware local EDT with value 1 within a 4 mm anchor radius, followed by linear decay to 0 at a 40 mm support radius; the repository release associated with this paper contains this corrected renderer. Correcting the encoding, with identical weights, raised the score from 0.40 to 0.7702. Because such a mismatch produces no error (the tensors are correctly shaped and the pipeline runs to completion), we suggest that interactive segmentation pipelines should include an explicit assertion that the guidance rendered at inference matches the training-time renderer, ideally verified end-to-end against ground truth on at least one case before submission.

A second observation concerns metric selection. The distinction between Dice computed only over lesion-bearing studies and Dice computed over all studies is substantial when a large fraction of studies contain no lesions, since a model that hallucinates on a healthy study scores zero there with no opportunity for compensating overlap. Development metrics that exclude empty-ground-truth cases can therefore give a considerably more optimistic picture than the challenge metric. We also found that optimising post-processing thresholds against voxel Dice alone was insufficient: retuning the same filter against a lesion-instance F1 criterion selected different thresholds, since a rule that deletes an entire connected component can be nearly neutral in voxel overlap while being decisive at the lesion level.

The main limitation of this work is compute. The submitted model was trained on a single fixed split and was not cross-validated, and the 0.058 Dice spread observed between two independently trained splits of the guided baseline suggests that some of the differences in our ablation table are within split-induced noise. The architecture comparison (+0.040) and the PET normalisation result

(-0.065) are large enough to be credible; the 0.0007 gap between the two distance encodings is not. Our postprocessing thresholds were likewise selected during preliminary development rather than by a held-out sweep, and their contribution should be read with that in mind.

5 Conclusion

We presented a round-supervised interactive segmentation model for whole-body PET/CT lesion delineation, in which corrective guidance is generated from the model’s own errors across successive simulated interaction rounds rather than from ground truth. Combined with a spacing-aware local distance encoding of user clicks and a residual-encoder 3D U-Net, this achieved 0.7702 Dice and 0.7558 lesion-level F1 on the autoPET V preliminary test set. Our ablations indicate that network capacity and the training-time interaction regime dominate the choice of guidance encoding, and we report negative results for organ-aware false-positive suppression and PET-specific normalisation. We additionally document a silent train/inference encoding mismatch that cost 0.37 Dice without producing any error, as a caution for interactive pipelines where guidance is rendered separately at training and deployment time.

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Table 1. Algorithm details

Team name	algorithm name (as submitted on grand-challenge)	data processing	pre-processing	data processing	post-training augmentation	data
Volyagers	autoPETV EDT+	CT clip 99.5%ile z-score; per-image z-score; resample to $3.0 \times 2.04 \times 2.04$ mm; clicks $\rightarrow$ spacing-aware local EDT (1 at click $\rightarrow$ 0 at 40 mm), unnor-malised	0.5+ PET z-score	Connected-component filter (18-connn.): re-move component only if volume < 0.35 mL <i>and</i> $SUV_{\max} < 6.0$	Rotation, scaling, Gaussian noise, Gaussian blur, brightness, contrast, low-resolution simulation, gamma, mirroring (nnU-Net defaults)	
test augmentation	time	ensembling (e.g. cross-validation, model ensemble, ...)	standardized framework? (e.g. nnUNet, MONAI, ...)	network architecture (e.g. UNet (3D))	ar-	loss
– (disabled)		– (single model, single fold)	nnU-Net (2.6.0)	v2 3D Residual Encoder U-Net; 6 stages; [32, 64, 128, 256, 320, 320]; blocks [1, 3, 4, 6, 6, 6]	En-	Dice + cross-entropy, deep supervision
training data	data/model dimensionality and size	use of trained els	pre-mod-	GPU hardware for training		
1611 FDG + PSMA PET/CT studies of autoPET (challenge data only; no external data)	3D: $112 \times 160 \times 128$ patch, batch size 2, 4 input channels (CT, PET, FG guidance, BG guidance)	– (trained from scratch)	(public available or own developed)	1 $\times$ NVIDIA H200; ≈ 97 h wall-clock to epoch 950		

Table 2. Ablation results. Fixed 250-case subset, 90 epochs, identical splits, one factor varied at a time. Mean Validation Dice from full sliding-window inference.

Factor	Variant	Mean Val. Dice
Guidance encoding	Truncated EDT	0.7102
Guidance encoding	Geodesic distance	0.7109
Guidance encoding	Disk	0.6957
Guidance encoding	Gaussian heatmap	0.6890
Normalisation	PET-specific normalisation	0.6457
Architecture	Plain U-Net	0.7102
Architecture	Residual encoder (L)	0.7506

Table 3. Preliminary test set results.

Configuration	Dice	F1
Baseline, single fold + TTA	0.7242	0.7002
Round-supervised interactive model	0.7702	0.7558